Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python** and **Java** services, production **AI-powered applications**, and data pipelines on cloud platforms, with PyTorch research experience, microservices, automated testing, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Advanced NLP, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Built **data pipelines** across PostgreSQL, SQL Server, and Azure storage with incremental loads and validation, sustaining 99%+ reliability over 5K+ monthly jobs, and cut dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built an **AI-powered application** on AWS that routes requests across GPT-4o, Claude, and Qwen-VL behind a switchable provider interface, serving 150+ instructors and 10K+ daily requests through React and Flask APIs.
- Designed multi-step **prompt chains** for rubric generation, multimodal vision OCR, and grading with fallbacks and per-call cost tracking, reaching 90%+ grading consistency and cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** for offline reinforcement learning, with end-to-end data pipelines processing 100K+ trajectory logs, improving long-horizon trajectory planning 18% in closed-loop simulation.
- Designed a multi-head **attention** encoder fusing map geometry with agent trajectories, benchmarking it against three baselines with Hydra and Weights & Biases and improving closed-loop planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented **Java** and Python **microservice** modules with Singleton and Factory patterns for cross-platform data sharing, covered by 20+ Mockito and PyTest suites in nightly Jenkins pipelines.
- Redesigned AWS Glue and S3 **ETL pipelines** and optimized APIs with pagination and caching, boosting pipeline efficiency 68% and response times 30% in Agile sprints with Microsoft and GitHub stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built Java Spring Boot **microservices** with RESTful APIs and real-time WebSocket updates for a resource-management platform, defining boundaries across 8 services and syncing 2M+ records across locations.
- Shipped features through Jenkins CI/CD with automated tests, Docker packaging, and Sentry and Azure Monitor **production monitoring**, while usage analytics guided fixes that cut resource wastage 42%.

Skills

Languages & Design: Python, Java, TypeScript, JavaScript, SQL, C++, Design Patterns
AI & ML: PyTorch, TensorFlow, Transformers, LLM Systems, Prompt Engineering, Embeddings, FAISS, Reinforcement Learning
Backend & Systems: FastAPI, Flask, Spring Boot, REST APIs, Microservices, Distributed Systems, Async Processing
Data & Cloud: PostgreSQL, SQL Server, MySQL, ETL Pipelines, Feature Engineering, AWS, Azure, GCP, Docker
Practices: Data Structures, Algorithms, Unit Testing, PyTest, Mockito, CI/CD, Jenkins, Git, Agile

Projects

Issue Tracker MCP Server | *Python, MCP, FastAPI, OAuth 2.0, PostgreSQL, LLM APIs*

- Designed an **MCP** server exposing issue search, page lookup, and ticket updates as typed tools with permission-scoped OAuth access, plus a **function-calling** agent that triages new tickets and an evaluation suite.

Duplicate Issue Detection Service | *Python, scikit-learn, PyTorch, Embeddings, FastAPI, Docker*

- Designed an **ML** service that embeds issue text, finds likely duplicates with nearest-neighbor search, predicts priority with a scikit-learn classifier, and serves results through a containerized **FastAPI** microservice with offline precision checks.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, Convex, GCP*

- Built an AI collaboration platform with LLaMA-2, transformer embeddings, and FAISS **semantic search** over 50K+ documents, adding Clerk auth and Convex real-time collaboration and cutting retrieval latency 40%.